

by the concept of 'bundling protocol suite.' The bundling protocol suite is believed to provide a general-purpose delay-tolerant protocol.

IPN proposes to interconnect IPN regions by way of common messaging layers called 'bundles.' Bundling is an overlay protocol that puts together a set of heterogeneous Internets. Bundles are arbitrarily long messages designed for end-to-end delivery between IPN nodes over distinct or identical transport layers. Delivery is always at bundle granularity.

The IPN will work more like e-mail. E-mails are sent over message switching, where information is transferred by store-and-forward process to any hub on the system and where the entire message is treated as one unit. The delay-tolerant IPN network will provide an always-on connection between planets, spacecrafts and the terrestrial Internet.

The store-and-forward technique of the IPN will help to minimise problems that crop up due to the vast distances involved, such as high error rates and latency periods that are minutes or even hours long (versus fractions of a second on the earth).

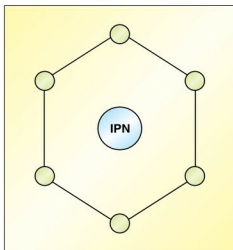


Fig. 1: Architecture of interplanetary Internet

Bundles (messages) may be fragmented and reassembled at the destination. These will be routed by a routing function through a concatenated series of Internets, like the terrestrial Internet protocol (IP) routes packets through a series of independent sub-networks. To achieve guaranteed end-to-end delivery, bundles may be retransmitted as in the automatic repeat request protocol used in the TCP layer of the terrestrial Internet.

The interplanetary backbone of the IPN depends on very fragile wire-

less links. Besides, the hubs on the interplanetary backbone are all moving with respect to each other. This may disturb the line-of-sight communication needing some sort of relay communication from time to time. Therefore unlike the earth's backbone environment of continuous connectivity, the IPN transmission may go for discontinuity.

Bundling protocol

As mentioned earlier, the IPN architecture will be based on an overlay protocol called 'bundling' that handles the complex environment of celestial communication. The functions of the bundling protocol are:

1. It operates in the message switching mode.

2. It operates in the custodial mode. The protocol does not wait for any acknowledgement from the receiver, and the intermediate nodes are given the responsibility to transfer bundles to the receiving station.

3. To tackle high error rate of the celestial link, hop-by-hop store-and-forward bundling with per-hop error-control mechanism is used.

4. Identification of the sender and the receiver in the IPN is done with an addressing scheme having names in two parts. Bundles are made with addresses or names of end points. The first part of the name is the routing part. This part gets the bundles for delivery to a remote destination called 'region' in the IPN. The second part of addressing is the administrative part. It contains the information that is essential to deliver bundles to one or more local destinations. Entity names are of the form '[admin part, routing part]'. The routing part is topologically significant. The admin part is transparent outside the region specified by the routing part. One example may be 'http://www.iste.org//mars.sol/'

Bundles are routed, forwarded and custody transferred with new concepts. In IPN, routing means selecting the best next IPN hop for the desired des-

Terrestrial vs Celestial Communication

Communication parameters	Wired terrestrial	Mobile ad hoc NET/ MANET/Wireless	IPN/Celestial/ Wireless
Power availability	Not critical	Important	Very crucial
Signal-to-noise ratio	Within acceptable range	Low	Very low
Error rate	Within acceptable range	Medium	High
Infrastructure	Defined/Fixed	Deployable	Deployable
Medium	Copper/Fibre	RF/IR	Primarily free space, RF
Delay in seconds	<1		10 to 10,000 (10,000 seconds means about 3 hours)
Deployment cost	Low	Medium	Very high
Operational cost	Low	Medium	Very high
Repairing terrestrial vs celestial communication and upgradation	Low	Medium	Very high